

Exploratory Data Analysis of Student Academic Performance

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ABSTRACT

The prediction and understanding of student academic performance are critical components of modern educational ecosystems. Educational institutions rely on continuous evaluation and data analysis to identify at-risk students and implement timely interventions. This project presents a comprehensive Exploratory Data Analysis (EDA) of a Student Performance Dataset containing records of 649 students and 33 distinct features.

The primary objective of this study is to uncover the underlying patterns, socio-demographic influences, and academic behavioural indicators that significantly impact the final grade (G3) of a student. Leveraging the robust capabilities of Python, Pandas, Matplotlib, and Seaborn, the analysis encompasses data inspection, statistical summarisation, and both univariate and bivariate visual analyses.

Key findings reveal that a student's prior academic history (G1 and G2 grades) is the most potent predictor of their outcome, exhibiting an overwhelming positive correlation (up to 0.92). Furthermore, behavioural traits such as dedicated study time show a positive correlation with higher grades, whereas high absenteeism and past academic failures negatively impact overall academic success. This report serves as a foundational analytical document that can be utilised to inform data-driven decision-making in educational administration and paves the way for advanced predictive machine learning models.

1. INTRODUCTION

In the era of big data, educational data mining has emerged as a pivotal field of research. Schools and universities generate vast amounts of data regarding student demographics,

social behaviours, academic backgrounds, and school-related activities. However, raw data alone does not yield actionable insights. It must be systematically processed, visualised, and analysed to extract meaningful narratives.

Exploratory Data Analysis (EDA) is the crucial first step in any data science pipeline. Coined by John Tukey, EDA is an approach to analysing datasets to summarise their main characteristics, often using visual methods. In the context of academic performance, EDA allows educators and data scientists to move beyond basic grade averages and understand the complex web of factors that contribute to student success or failure.

This project focuses on a dataset comprising 649 student records. By rigorously examining 33 variables—ranging from parental education and study habits to alcohol consumption and health status—this report aims to isolate the defining variables that dictate the final grade (G3). The insights derived from this analysis are intended to foster a deeper understanding of student dynamics and assist educational institutions in designing targeted support systems.

2. OBJECTIVES

The primary objectives of this project are strictly defined as follows:

1. **Data Familiarisation:** To systematically load, clean, and inspect the Student Performance dataset to ensure data integrity.
2. **Statistical Profiling:** To generate descriptive statistics that establish the baseline metrics for age, grades (G1, G2, G3), and other numerical features.
3. **Distribution Analysis:** To perform univariate analysis on the target variable (Final Grade - G3) to understand its overall distribution.
4. **Relationship Discovery:** To conduct bivariate analysis to evaluate how independent features (Study Time, Absences, Gender) influence the dependent variable (G3).
5. **Correlation Mapping:** To generate a correlation matrix to mathematically quantify the relationships between continuous variables, particularly past and future academic performance.
6. **Insight Generation:** To synthesise the analytical findings into actionable educational insights.

3. DATASET DESCRIPTION

The dataset utilised for this analysis is the **Student Performance Dataset (student-por.csv)**, commonly sourced from educational data mining repositories (e.g., UCI Machine Learning Repository). It details student achievement in secondary education for two Portuguese schools.

- **Total Records (Rows):** 649
- **Total Features (Columns):** 33

Key Attributes Analysed:

- **Target Variable:**
 - G3: Final grade (numeric: from 0 to 20).
- **Academic Attributes:**
 - G1: First period grade (numeric: from 0 to 20).
 - G2: Second period grade (numeric: from 0 to 20).
 - studytime: Weekly study time (numeric: 1 = <2 hours, 2 = 2 to 5 hours, 3 = 5 to 10 hours, 4 = >10 hours).
 - failures: Number of past class failures (numeric: n if $1 \leq n < 3$, else 4).
 - absences: Number of school absences (numeric: from 0 to 93).
- **Demographic & Social Attributes:**
 - age: Student's age (numeric: from 15 to 22).
 - sex: Student's sex (binary: 'F' - female or 'M' - male).
 - Medu / Fedu: Mother's and Father's education levels.
 - health: Current health status (numeric: from 1 - very bad to 5 - very good).
 - Dalc / Walc: Workday and weekend alcohol consumption (numeric: from 1 - very low to 5 - very high).

4. METHODOLOGY

The project methodology follows a structured Data Science life cycle tailored for EDA:

1. **Environment Setup:** Initialising the Python environment and importing essential libraries, including pandas for data manipulation, and matplotlib.pyplot and seaborn for data visualisation.
2. **Data Loading and Inspection:** Loading the student-por.csv file using Pandas. Executing commands like .head(), .info(), and .isnull().sum() to check for missing values, data types, and structural anomalies.
3. **Statistical Summary Generation:** Utilizing the .describe() function to calculate measures of central tendency (mean, median) and dispersion (standard deviation, min, max) for numerical columns.
4. **Univariate Analysis:** Creating histograms and density plots to analyze the standalone distribution of the final grade (G3).
5. **Bivariate Analysis:** Utilizing box plots, bar charts, and scatter plots to compare single predictor variables against the final grade to observe trends, medians, and outliers.
6. **Correlation Analysis:** Calculating the Pearson correlation coefficient across the dataset and visualizing the mathematical relationships using a comprehensive heatmap.
7. **Insight Extraction:** Translating visual and mathematical data into professional, readable conclusions.

5. STATISTICAL SUMMARY

Before visualizing the data, a rigorous statistical summarization was performed to establish

baseline metrics for the student population.

Student Age Metrics

- **Mean Age:** 16.74 years
- **Minimum Age:** 15 years
- **Maximum Age:** 22 years
- *Interpretation:* The dataset primarily consists of late-teen high school students, representing a standard secondary education demographic.

Academic Performance Metrics

Final Grade (G3):

- **Mean:** 11.91
- **Median:** 12.0
- **Minimum:** 0
- **Maximum:** 19
- *Interpretation:* The average performance hovers around 11.91 out of 20. The presence of a minimum grade of 0 indicates either severe academic failure or a high rate of absenteeism/dropouts resulting in a zero score.

Second Period Grade (G2):

- **Mean:** 11.57
- **Maximum:** 19

6. VISUALIZATION ANALYSIS

The core of Exploratory Data Analysis relies on visual storytelling. The following sections detail the visualizations generated during the project and the insights derived from them.

6.1 Grade Distribution Analysis

[Insert Figure 1: Grade_Distribution.png here]

(Placeholder for a Histogram/KDE Plot showing the distribution of the G3 variable)

Detailed Analysis:

Figure 1 illustrates the frequency distribution of the final grade (G3) across the entire student population. The visualization reveals that the majority of students achieved scores falling within the 10 to 15 mark range. The shape of the distribution is approximately normal (bell-shaped), which is typical for academic grading systems where the bulk of the population achieves average proficiency. Notably, the tails of the distribution are relatively thin, indicating that there are very few extremely low performers (excluding the anomaly at exactly 0, which often represents students who missed the final exam) and very few exceptional performers scoring near the maximum of 19 or 20.

6.2 Impact of Study Time on Final Grades

[Insert Figure 2: Studytime_vs_Grade.png here]

(Placeholder for a Box Plot or Bar Chart mapping 'studytime' categories against 'G3')

Detailed Analysis:

Figure 2 examines the bivariate relationship between reported weekly study time and the final grade achieved. The data categorizes study time into four distinct levels. The visualization demonstrates a clear upward trend: students who dedicate more hours to studying (Levels 3 and 4, representing >5 hours a week) generally achieve higher final grades compared to their peers in Levels 1 and 2. The median grade visibly increases as the study time bracket increases. This confirms the educational hypothesis that deliberate academic effort and increased study duration have a direct, positive impact on student success and mastery of the curriculum.

6.3 Effect of Absenteeism on Academic Success

[Insert Figure 3: Absences_vs_Grade.png here]

(Placeholder for a Scatter Plot or Box Plot comparing 'absences' with 'G3')

Detailed Analysis:

Figure 3 maps the number of school absences against the final grades. The visualization highlights a distinct trend regarding student attendance. Students exhibiting low absenteeism form a dense cluster around the higher grade brackets, indicating that regular attendance generally results in better academic performance. Conversely, as the number of absences increases, there is a noticeable dispersion and a downward trend in grades. Excessive absences are strongly associated with lower final grades, highlighting that physical presence and continuous engagement in the classroom are vital prerequisites for academic achievement.

6.4 Gender-based Performance Comparison

[Insert Figure 4: Gender_vs_Grade.png here]

(Placeholder for a Box Plot, Violin Plot, or Bar Chart comparing 'sex' ('M'/'F') against 'G3')

Detailed Analysis:

Figure 4 provides a demographic breakdown comparing the academic performance of male and female students. The visual analysis indicates that, on average, female students achieved slightly higher final grades compared to their male counterparts within this specific dataset. While the median grade for females is marginally elevated, the interquartile ranges and overall spread of the data suggest that the performance gap is not drastically wide. Both genders share similar minimum and maximum ranges, but the central tendency leans slightly in favor of the female demographic.

7. CORRELATION ANALYSIS

To move beyond visual trends and quantify the relationships between numerical variables, a Pearson Correlation analysis was conducted.

[Insert Figure 5: Correlation_Heatmap.png here]

(Placeholder for a Seaborn Correlation Heatmap displaying the matrix of numerical attributes)

Detailed Analysis and Interpretation:

Figure 5 presents a correlation heatmap, which uses color gradients to represent the strength and direction of linear relationships between variables. The most striking observation from this heatmap is the exceptionally strong positive correlation between the periodic grades.

- **G2 and G3 Correlation:** 0.92
- **G1 and G2 Correlation:** 0.86
- **G1 and G3 Correlation:** 0.83

Interpretation: An astronomical correlation of 0.92 between G2 and the final grade (G3) mathematically proves that previous academic performance is the strongest and most reliable predictor of future academic outcomes. A student who performs well in the second period is almost guaranteed to perform well in the final evaluation.

Negative Relationships: Furthermore, the heatmap reveals a notable negative correlation between past class **failures and the final grade (-0.39)**. This indicates a compounding effect of academic struggles; students who have failed in the past are statistically more likely to struggle and achieve lower grades in their current evaluations.

8. KEY FINDINGS

Synthesizing the statistical summaries, univariate distributions, bivariate visualizations, and correlation matrix, the following key findings have been established:

1. **Prior Performance is Paramount:** The strongest predictor of a student's final grade (G3) is their performance in preceding periods (G1 and G2). The correlation of 0.92 between G2 and G3 suggests minimal dramatic shifts in student performance near the end of the academic year.
2. **The Penalty of Past Failures:** A history of academic failure has a quantifiable negative relationship (-0.39) with current success, suggesting that struggling students remain at risk without targeted intervention.
3. **Study Effort Yields Results:** There is a definitive positive impact of increased weekly study time on final academic outcomes. Students dedicating more than 5 hours a week show noticeably higher median grades.
4. **Attendance Matters:** High absenteeism is detrimental to academic success. Lower attendance records consistently map to poorer final grades.
5. **Behavioral Impacts:** The analysis indicates a slight negative relationship between frequent alcohol consumption (both workday and weekend) and overall grade achievement.
6. **Normal Distribution of Success:** The grading curve is healthy, operating on a normal distribution primarily centralized between the scores of 10 and 15, with very few statistical outliers.

9. CONCLUSION

This Exploratory Data Analysis project successfully parsed the Student Performance Dataset to uncover the hidden dynamics of academic success. By utilizing Python-based data science libraries, we transformed 649 rows of raw data into a coherent narrative.

The analysis empirically validates several educational hypotheses: consistent study habits and regular attendance are foundational to achieving high grades. More importantly, the data proves that academic performance is highly continuous; early struggles (low G1/G2 scores, past failures) almost inevitably lead to poor final outcomes. Therefore, the EDA successfully identified the major contextual factors affecting student success, providing a clear roadmap for where educators should focus their attention to improve student outcomes.

10. FUTURE SCOPE

While this EDA provides profound descriptive insights, the dataset holds immense potential for predictive analytics. Future scopes of this project include:

- **Predictive Modeling:** Developing Machine Learning models (such as Random Forest Regressors, Support Vector Machines, or Linear Regression) to predict a student's final G3 grade based on their early semester data and demographic profile.
- **Classification Systems:** Creating a binary classification model to flag students as "At-Risk" or "Safe" regarding passing the academic year, allowing for early academic intervention.
- **Feature Engineering:** Combining social metrics (like alcohol consumption, parental education, and free time) into an overall "Socio-Environmental Score" to better measure

out-of-school impacts on learning.

- **Broader Datasets:** Applying the same analytical pipeline to larger, multi-school datasets to ensure the findings are universally applicable rather than localized to a specific demographic.

11. REFERENCES

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